What tech do we need?

Michael Winkelmann

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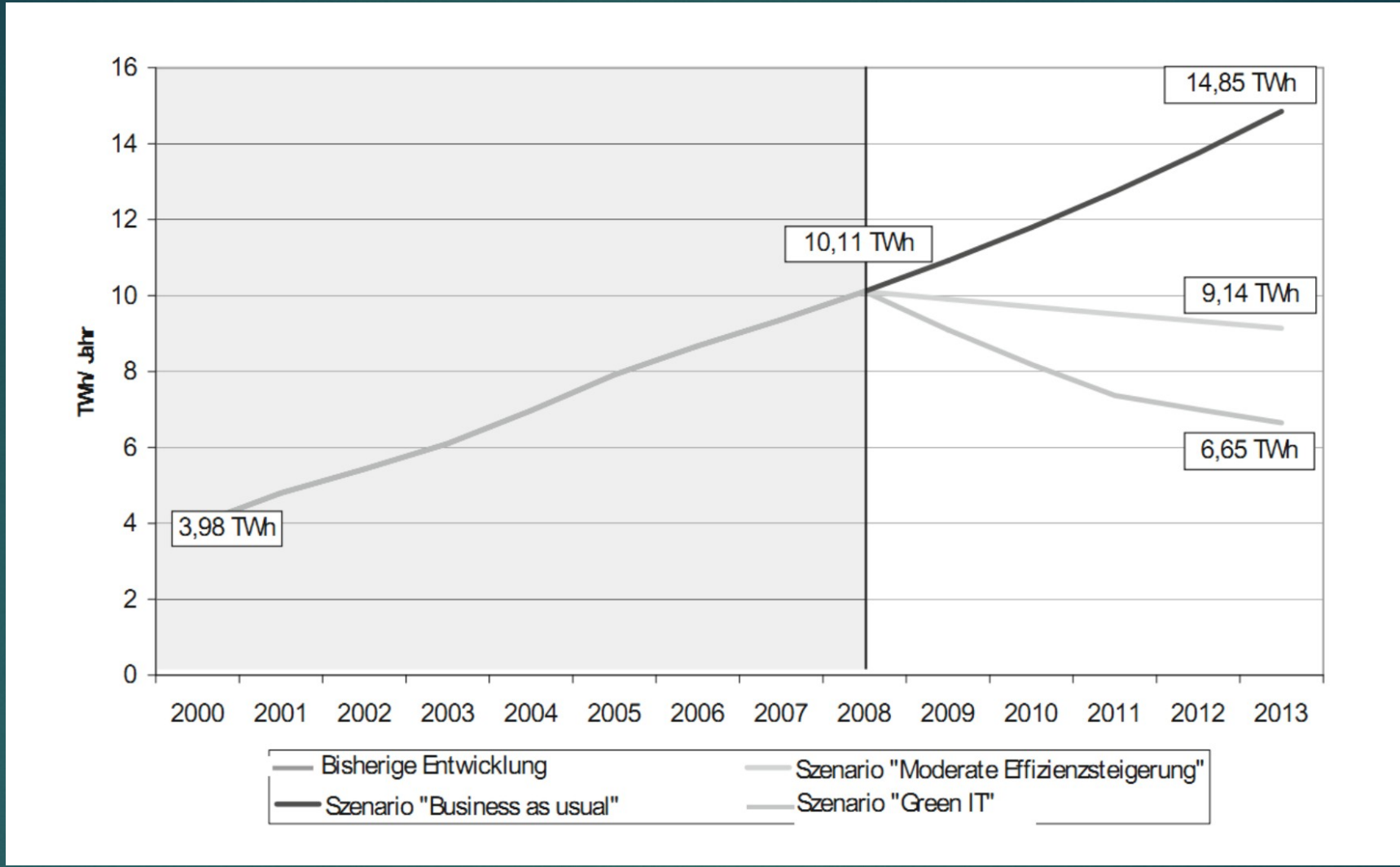
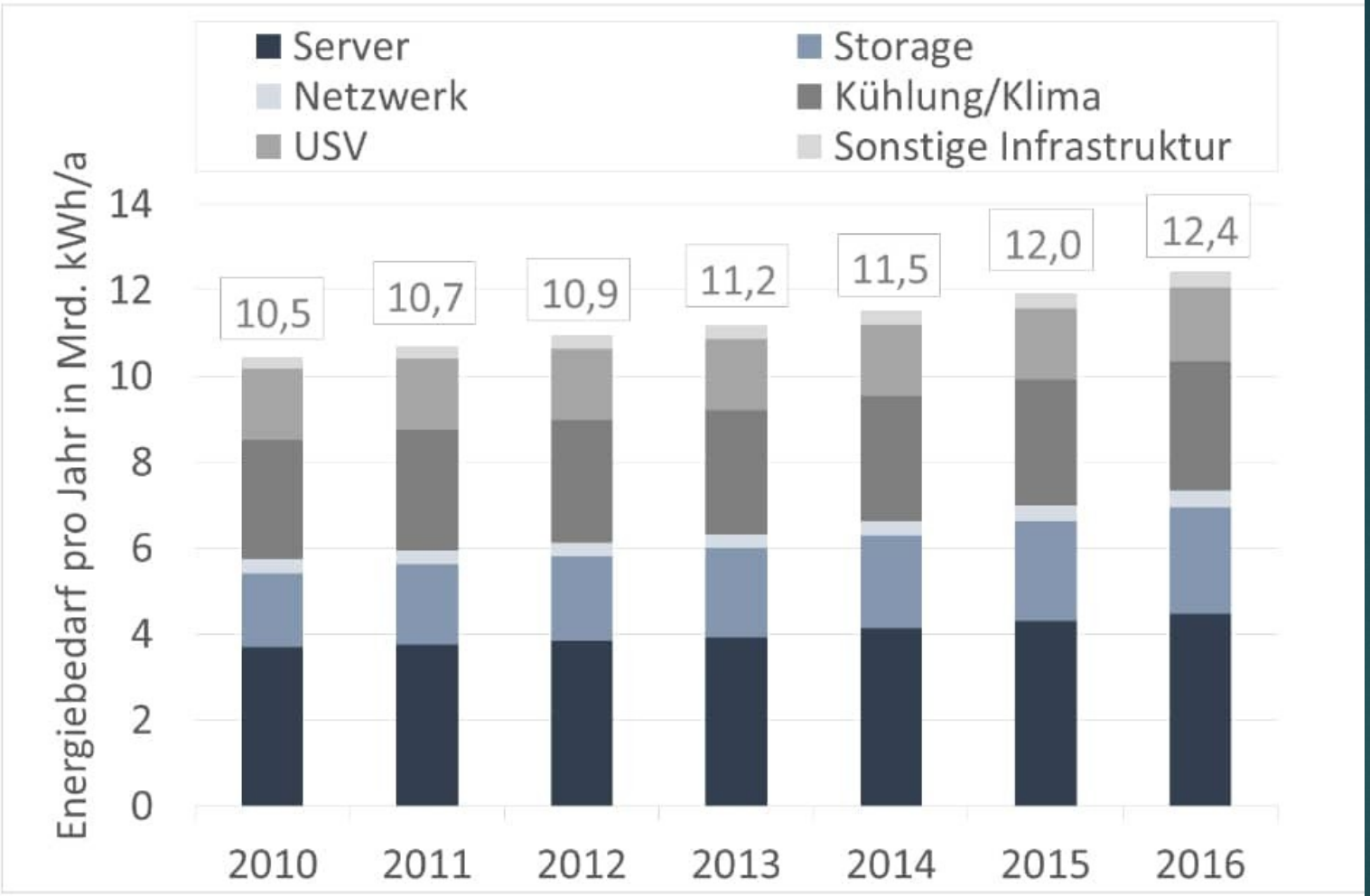


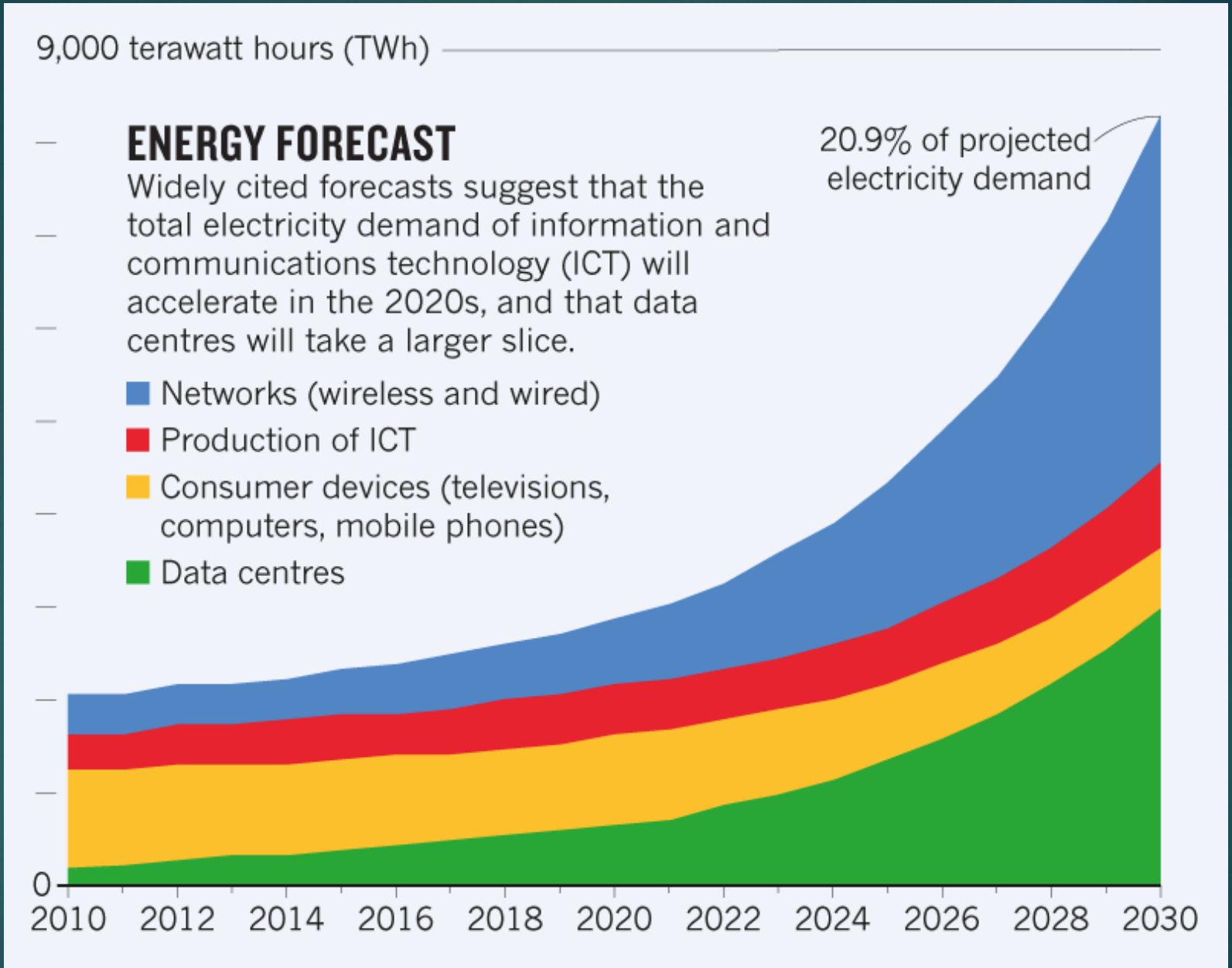
FICHTER, Dr. K. ; BEUCKER, Dr. S. ; CLAUSEN, Dr. J. ; HINTEMANN, Dr. R.: Green IT: Zukünftige Herausforderungen und Chancen / Borderstep Institut für Innovation und Nachhaltigkeit gemeinnützige GmbH, 2009. – Forschungsbericht

Abbildung 7: Entwicklung des Stromverbrauchs der Rechenzentren in Deutschland in den Jahren 2010 bis 2016



Quelle: Borderstep 2017 (Hintemann, 2017a)

https://www.greenit-solution.ch/energieeffizienz-im-rechenzentrum



https://www.nature.com/articles/d41586-018-06610-y

Sustainability

	Efficiency	Consistency	Sufficiency
	"We need better tech!"	***"We need wasteless tech!"***	***"We need less tech!"***
	• More FLOPS/Watt • More Requests/Sec • Sending less data • Virtual Servers	• Use heat from data centers for warm water and heating • Recyclable Hardware	• Renunciation • Reduce • De-growth
Potential*	10-20%	50-80%	10-40%
	• Rebound effects	• Hardware is hard to recycle	• Almost only discussed in non-tech groups

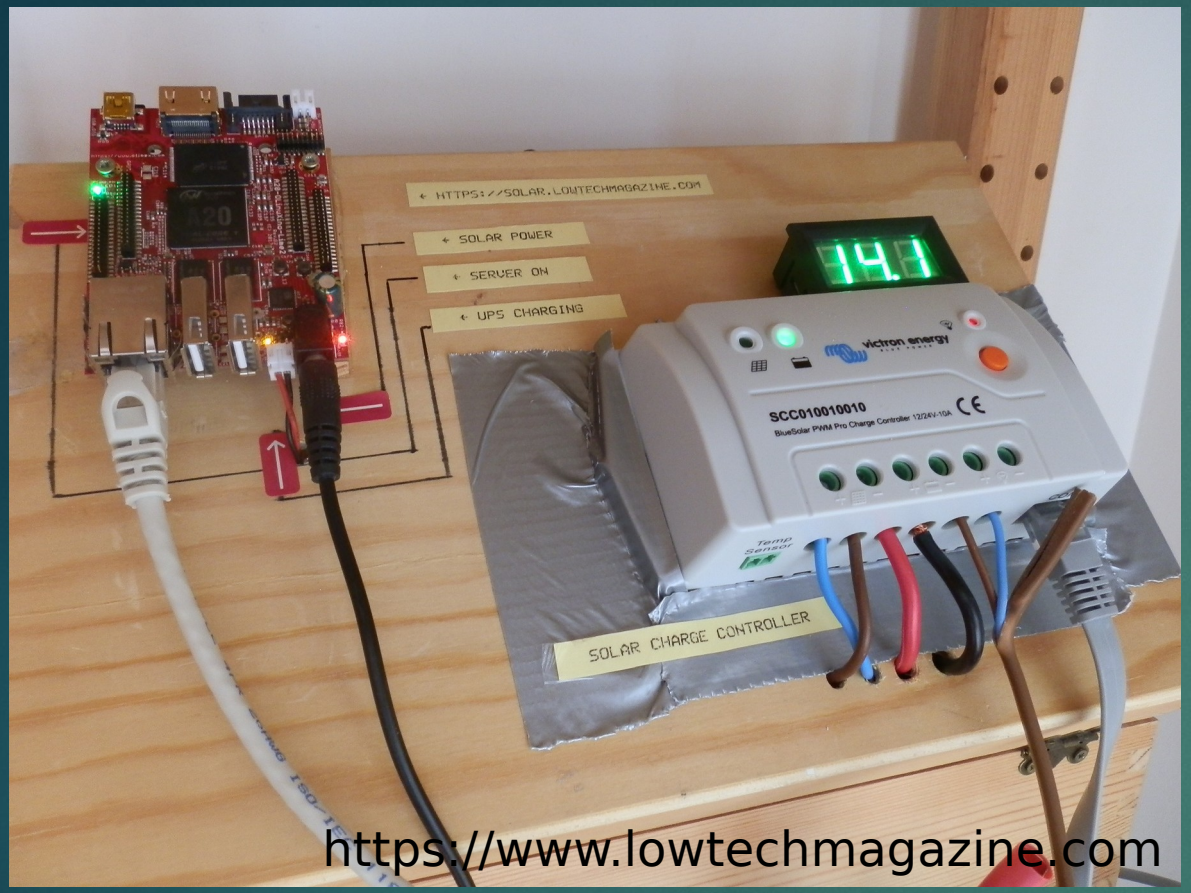
*https://evolution2green.de/sites/evolution2green.de/files/documents/evolution2green_inputpapier_effizient_konsisten_suffizienz.pdf

http://sites.psu.edu

https://www.amishgazebos.com

Amish: Sustainable way of living through sufficiency?

Solar-powered Webserver



Tiny House Movement

...Small is beautiful?

What tech do we **<u>NOT</u>** need?

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Greentechfestival.com
“#celebratechange”

Aquon One

Freedom by sunlight

Automobili Pininfarina

The fastest car in the world

Volocopter

Urban. Autonomous. On demand.

Sonos to deny software updates to owners of older equipment

Speakers and networks will miss out on security fixes and eventually stop working

This $1,500 Toaster Oven Is Everything That’s Wrong With Silicon Valley Design

Automated yet distracting. Boastful yet mediocre. Confident yet wrong.

Seattle startup Nori raises cash for blockchain marketplace to fight climate change

Number of In-Vehicle Displays to Increase Dramatically in 2020s

“OEMs are asking us to support up to 12 screens in future cars, for both the driver and passengers. And these screens won’t just be HD, they’ll be 4K, spread all around the car.”

https://www.tu-auto.com

Summary and further actions

- Sufficiency is mostly ignored as a topic in the tech & sustainability debate
- Evaluation of recent technology innovation in terms of sustainability is needed ("*Enkeltauglichkeit*")
- Peer-reviewed studies about technologies and products
- Open a slack channel to collect topics and discuss
- A popular format for Tech Evaluation, like a short videoclip format or podcast
- Tell a better story for a sustainable lifestyle